

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

Sebastian Bobadilla-Suarez
Independent Researcher
`sebastian.bobadilla.s@gmail.com`
ORCID: 0000-0002-5951-0772

20 September 2026

Abstract

Collective phase order can coordinate distributed cortex, but does not establish shared content or self-representation. We distinguish coherence, compatibility and accurate reconstruction. Shared content requires independently identified quantities on overlapping regions; compatible local descriptions determine a unique global state. A contraction fixes that state uniquely if it reconstructs the state exactly; fidelity across other states separately requires discriminating codes. Counterexamples show why a phase resultant and fixed-point uniqueness cannot replace these measurements. Lean 4 verifies the conditional implications and a model jointly satisfies their eight connecting hypotheses. This establishes mathematical consistency, not cortical realization. A declared actuator's stored mode energy bounds mean coupling and yields a necessary condition for a coherent stationary branch; cortical mode prices remain unknown. We propose a held-out comparison of field-score and structural-connectivity onset models during anaesthetic emergence. A predictive advantage would support the specified spatial model; causal field identification requires intervention. Unity and a minimal self remain interpretations without an independent measure of experience.

1 Introduction

Endogenous extracellular fields can affect spike timing and may coordinate activity across cortical regions (McFadden, 2002, 2020; Pockett, 2012; Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Coordination raises a further question: when do those regions describe one situation, and when does the resulting state represent its own place in it?

Consider reaching for a cup. Vision and movement can describe different features while agreeing on the cup’s location. Matching their timing leaves agreement about that shared quantity untested. An internal description of the perceiver’s relation to the cup asks for still more. We call the three requirements *coherence* (collective phase order), *compatibility* (agreement of local descriptions on overlaps), and *self-representation* (accurate reconstruction from an internal encoding). The mathematical states are spatial measures and fields; identifying them with perceptual content requires independent decoding evidence.

The familiar noisy mean-field oscillator supplies a stationary coherence threshold (Sakaguchi, 1988; Strogatz & Mirollo, 1991). We add a resource condition for a declared actuator, bounds that connect phase order to content only through specified encoders, and conditions under which compatible descriptions glue and an accurate reconstruction fixes the glued state. A Lean 4 development checks these implications and a joint witness satisfies their eight connecting hypotheses in a specified model. It proves neither that

23 cortex satisfies those hypotheses nor that the glued state is experienced. The
24 latter identification motivates the framework and needs its own endpoint.

25 The empirical question is which measurements could support the pro-
26 posed connections. A phase resultant can miss a locally ordered winding field;
27 stationary phase distributions can leave the coupling mechanism unidenti-
28 fied; and coherent adaptation can lose alignment with a designated content
29 template. Independent decoders must identify shared quantities before their
30 agreement can be tested. We therefore distinguish stationary branches from
31 finite observed onset, and compare prespecified field and connectivity onset
32 models on held-out emergence recordings. The comparison tests a specified
33 spatial prediction, not the presence of experience.

34 Sections 2–5 develop the three requirements; Section 6 applies their mea-
35 surement obligations to a digital candidate. Section 7 states the resource and
36 composition premises, and Sections 8–9 examine observables and the cortical
37 test. Table 1 lists the eight connecting hypotheses; Supplemental Material
38 gives their derivations and controls.

39 **2 Limits of phase order**

40 Several limits hold without the cortical hypothesis. Equal phases can coexist
41 with incompatible local descriptions; phase order constrains shared content
42 only through encoders related on overlaps. A winding field can have zero
43 global resultant while remaining locally ordered. A constant readout can

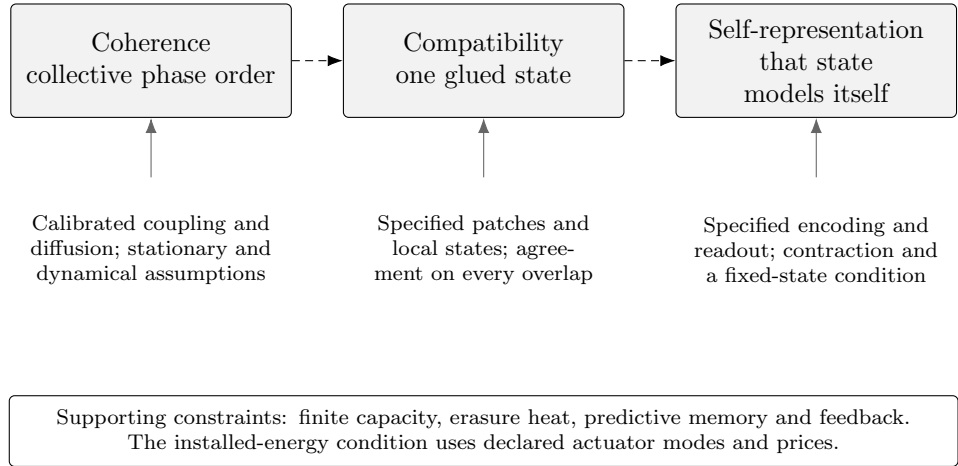


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

44 have a unique fixed point without reconstructing its input. A region that
45 does not distinguish relevant states cannot encode them, and a site cannot
46 report a change outside its causal past by a declared deadline. These lim-
47 its reflect familiar phase oscillator and information constraints (Acebrón et
48 al., 2005; Ott & Antonsen, 2008; Dörfler & Bullo, 2014; Breakspear, 2017).
49 The sections below state each one beside the condition or measurement it
50 motivates.

51 **3 Coherence: physical coupling and its re-** 52 **source condition**

53 **3.1 Dynamics and the stationary threshold**

54 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

55 Here θ_i is the phase of oscillator i , ω_i its natural frequency, $D > 0$ is phase dif-
56 fusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units,
57 with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; fre-
58 quency heterogeneity and diffusion are distinct parameters. The stationary
59 calculation below uses identical frequencies and this scalar coupling. Ap-
60 plying it to a spatial kernel requires an aggregation rule and a check of the
61 resulting approximation.

62 For identical frequencies in their rotating frame, the positive normalized
63 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

64 where I_n are modified Bessel functions and ψ is the collective phase. The for-
65 mal derivation uses the absence of stationary current for a periodic gradient
66 drift to obtain this density, then verifies normalization and self-consistency.

67 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988; Strogatz &
68 Mirollo, 1991). Above threshold, exactly one positive solution coexists with
69 zero and increases strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

70 Figure 2 shows this stationary branch.

71 Linearization about the uniform density gives rate $K/2 - D$ for the first
72 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists
73 exactly above threshold. About the coherent density, the quadratic form
74 is strictly dissipative transverse to collective rotation, whose rate is zero.
75 These linear estimates concern the limiting equation. Relating them to finite
76 stochastic trajectories requires a dynamical mean-field limit and branch selec-
77 tion; exchange symmetry alone does not establish the needed independence
78 (Sznitman, 1991).

79 **3.2 Installed energy bounds mean coupling**

80 A spatial model replaces the finite sum by integration against a kernel. One
81 declared implementation carries response modes with continuous profiles ϕ_i ,
82 installation prices u_i and nonnegative occupancies $c_i(\Xi)$ in configuration Ξ :

$$K_\Xi(x, y) = \sum_i c_i(\Xi) \phi_i(x) \phi_i(y), \quad U(\Xi) = \sum_i u_i c_i(\Xi). \quad (4)$$

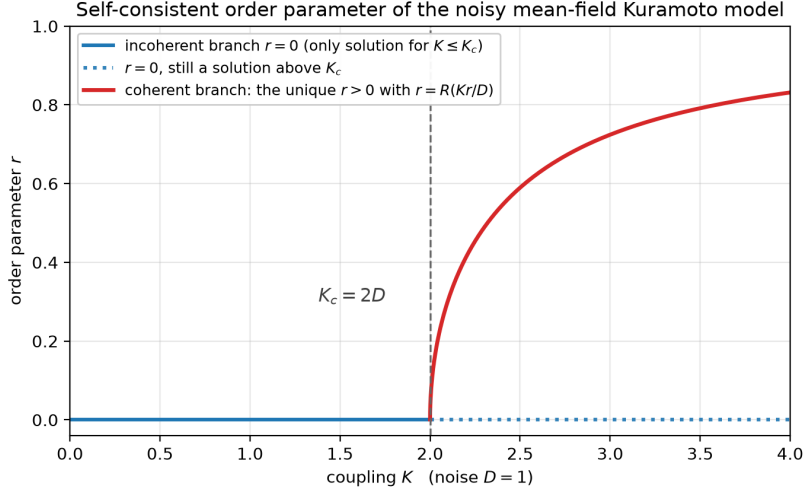


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

83 The same configuration sets the reciprocal kernel and stored energy. An
 84 executed transition changes the kernel through its changed occupancies and
 85 requires work $U(\Xi') - U(\Xi) + Q$, with Q positive into the reservoir. The
 86 installed kernel is averaged over the resulting configuration law $p(\Xi)$. Cell-
 87 pair reconstruction on refining spatial partitions converges to its product-
 88 measure integral under the specified compactness and sampling assumptions;
 89 the refinement index is independent of the number of control updates.

90 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 91 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's

92 integral gives

$$\begin{aligned}
 U_{\text{inst}} &= \sum_{\Xi} p(\Xi) U(\Xi), \\
 K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_{\Xi} p(\Xi) \sum_i c_i(\Xi) m_i^2 \leq \kappa U_{\text{inst}}.
 \end{aligned}
 \tag{5}$$

93 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 94 coherent stationary branch and any growing Fourier mode about the uniform
 95 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 96 is necessary: energy above it need not produce sufficient coupling because
 97 the bound can be loose.

98 The theorem concerns stored energy in the declared actuator modes. The
 99 estimated cortical signalling expenditure and extracellular electrostatic en-
 100 ergy in Supplemental Material measure different quantities; neither identifies
 101 the occupancies, mode prices or their relation to metabolic supply. Conse-
 102 quently the bound supplies a necessary condition for a declared actuator,
 103 but no cortical value of κ or demonstration that cortex meets it. Applying
 104 the scalar threshold to heterogeneous or signed spatial dynamics also needs
 105 a justified reduction.

106 4 Compatibility: from coordinated activity 107 to a global description

108 4.1 Why common timing is insufficient

109 For overlapping patches U and V , compatibility means that restricting either
110 local description to $U \cap V$ gives the same object. Descriptions can agree on
111 shared quantities while contributing different information elsewhere. Com-
112 patible sections of the specified sheaf have a unique global extension. On
113 a finite cover, local finite spatial measures agreeing as measures on every
114 overlap also have a unique finite-measure extension whose sheaf image is the
115 glued section. No mass-one normalization is imposed.

116 In the cup example, visual and reaching descriptions must express a
117 shared quantity, such as cup location, in common coordinates. Restriction
118 retains that quantity while dropping details particular to each description.
119 Agreement permits surface appearance and hand configuration to contribute
120 to one situation. The mathematical counterexample assigns the same phase
121 to every patch while two local mass profiles disagree on a shared site. Those
122 profiles have no common extension. Because they are chosen independently
123 of phase, synchrony places no constraint on their disagreement.

124 This distinction limits the biological interpretation. Receptive territories
125 and content domains must be related to spatial patches, and the descriptions
126 must support the required extension. Matching overlap marginals of de-

127 coded probability laws need not determine a unique global joint distribution
 128 (Abramsky & Brandenburger, 2011).

129 The cover is itself a choice, and the construction constrains it in one way
 130 only: synchronized patches agree where they overlap. Where they do not
 131 overlap there is nothing to compare, so a cover by pairwise disjoint patches
 132 satisfies that obligation for *every* assignment of local descriptions and phases
 133 and the gluing theorem still returns a unique global section, which is the
 134 tuple it was handed. Membership of the class then reports nothing about the
 135 system beyond the patches covering it. What a gluing claim asks of a cover
 136 is therefore a non-degeneracy the theorem does not assume: that distinct
 137 patches meet, and that what they share is a quantity fixed independently of
 138 the agreement it is being used to establish.

139 4.2 When coherence does constrain content

140 A declared relation between phase and content gives a positive result. Let
 141 contents take values in a real normed vector space $\mathcal{C}$. Patch i reads an
 142 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 143 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 144 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 145 sites; their equality on overlaps is a supplied relation between descriptions of
 146 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 147 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every

148 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1 - r^2}. \quad (6)$$

149 Here discrepancy is the norm of the difference, and r and N are read on
150 whichever population the bound is applied to. That choice is what decides
151 whether the estimate says anything. A patch carries an order parameter of
152 its own, and reading the bound there carries that patch's resultant and that
153 patch's site count, so a large population observed through small patches is
154 constrained by the small count. Read instead on the whole array, the factor
155 N makes the estimate weak, because it draws a pointwise guarantee from
156 an average over N^2 pairs. At $r = 1$ the observations agree exactly at either
157 scale. A shared scalar encoder is a special case. Dropping the overlap rela-
158 tion permits disagreement even at perfect locking. A common nonexpansive
159 linear projection preserves agreement, but agreement of a projection does
160 not establish agreement of the full vector.

161 What Eq. (6) bounds is a chord, which never exceeds 2, so the estimate
162 constrains anything only where the right-hand side falls below that. At
163 the sheet's own patch order, the bound is informative for at most 3 sites;
164 beyond that, the coherence-to-content link requires the bridge assumption
165 (E78) of the cortical hypothesis. Supplemental Material provides the detailed
166 derivation of this saturation, and states three conditional derivations that
167 replace the uniform factor N by a count of disagreeing pairs, a chain across

168 a cover's shared sites, or a spectral gap at a frequency-locked configuration.

169 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 170 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 171 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 172 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

173 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 174 duce an initial mismatch. Approximate observations leave a nondecaying
 175 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 176 update to the phase process.

177 4.3 What approximate agreement determines

178 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 179 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 180 Weights that vanish outside their patches and sum to one at each site select
 181 a global profile by weighted averaging. The selection lies within ε of each
 182 patch there, and any two such selections differ by at most ε , independently
 183 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
 184 $L\varepsilon$ across selections.

185 At zero tolerance, the selection is the unique exact extension. At every
 186 positive tolerance there are families with no exact extension; selection then
 187 gives an approximation whose dependence on weights is bounded. Supple-

188 mental Material gives a three-site witness attaining the bound. The mass-
 189 profile construction uses the scalar case of the content dynamics and addi-
 190 tionally requires nonnegativity of the initial profiles and observations. The
 191 general vector result supplies no identification with the spatial-measure sheaf.

192 These constructions concern declared objects: a specified cover, contents
 193 valued in a normed space, and encoders related on overlaps. They select
 194 neither the neural cover nor the decoder, and identifying their contents with
 195 perceptual content, or establishing that the identification bears on experi-
 196 ence, is a further modelling task; Section 8 sets out what a measurement of
 197 it would have to establish.

198 **5 Self-representation: a state that models it-** 199 **self**

200 **5.1 The reflexive construction**

201 Compatibility gives a global state. Reflexivity relates it to an encoding within
 202 that state. Equip global sections with a complete metric. If the additional
 203 restriction-resonance condition identifies encoding with restriction to a region
 204 $\mathcal{A}$, the reconstruction map is

$$F(s) = \text{readout}(s|_{\mathcal{A}}). \tag{8}$$

205 The region and readout are fixed across the state space. If F contracts and
 206 fixes the section s_* formed from the local descriptions, Banach's theorem
 207 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89 sup-
 208 plies the contraction and $F(s_*) = s_*$; restriction resonance is separate. Su-
 209 percritical coupling makes the stipulated contraction factor $\Lambda = e^{-(K-2D)\tau/2}$
 210 less than one for a relaxation time $\tau > 0$. Constructing a map with that
 211 factor remains a modelling obligation.

212 In the cup example, the global state includes the represented scene and the
 213 perceiver's relation to it. A local encoding must retain enough information
 214 for the readout to recover that modelled state; cup position alone would be
 215 insufficient if hand configuration and viewpoint could not be reconstructed.

216 For the same map with Lipschitz factor $\Lambda < 1$, approximate reconstruc-
 217 tion gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - \Lambda}. \quad (9)$$

218 At the stipulated rate this radius grows as $2\varepsilon/((K-2D)\tau)$ near threshold, for
 219 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
 220 weakens.

221 **5.2 Reconstruction across a family of states**

222 Sensitivity can be required quantitatively. Declare relevant macrostates, an
 223 encoder Φ and readout R , and require $d(s, R(\Phi(s))) \leq \varepsilon$ throughout that

224 family. If two states have the same code, the triangle inequality gives

$$\Phi(s) = \Phi(t) \implies \max\{d(s, R(\Phi(s))), d(t, R(\Phi(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

225 Thus M relevant states pairwise more than 2ε apart require at least M
226 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
227 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
228 existing self-prediction map.

229 This is an additional accuracy criterion on a family, with no requirement
230 for multiple exact fixed points. A blind encoder or constant readout fails it
231 for separated states despite having a unique fixed point.

232 Where the encoding is read as the state's own restriction to a region, the
233 criterion becomes a condition on the region. Restriction resonance requires
234 the encoder and that restriction to identify the same pairs of states, so a
235 boundary reconstructing two relevant states more than 2ε apart while its
236 region reads them identically is not resonant. Read forwards, a resonant
237 boundary that reconstructs a pairwise separated family restricts injectively
238 on it. Which states a region reads identically is fixed by where the region
239 sits, before an encoding is chosen, so the resolution the family demands is
240 the region's and no readout supplies it. The implication runs one way: a
241 region that does resolve the family is not thereby shown to carry a resonant
242 encoding, and E89 supplies no such resolution.

243 Those two demands are jointly satisfiable with nothing folded anywhere.

244 Take the region to be the whole substrate: restriction is then the identity, so
 245 the boundary is resonant, reconstructs every declarable family exactly, and
 246 restricts injectively on all of them. A properness condition on the region is
 247 therefore a separate requirement rather than a consequence of resonance and
 248 resolution, and what a fold into a local region is asked for. Its quantitative
 249 form replaces the alphabet by a metric. With an L -Lipschitz readout, a
 250 family reconstructed within ε whose members lie pairwise more than $d_{\min}$
 251 apart has δ -separated codes whenever $\delta L \leq d_{\min} - 2\varepsilon$, so the family is no
 252 larger than the δ -packing number of what the region carries. That is the
 253 shape a continuous family of states needs, and at the whole substrate it says
 254 nothing, because the region's range is everything. The gain L is declared and
 255 derived here for no physical readout, and at unbounded gain the bound says
 256 nothing.

257 **5.3 Agreement after a change takes communication**

258 Snapshot compatibility does not specify how quickly descriptions respond
 259 when a shared quantity changes. In a synchronous network whose updates
 260 read only the site's own state and declared neighbours' messages, two initial
 261 configurations agreeing on a site's causal past through T rounds produce the
 262 same state there at round T . The deadline consequence is the one Section 2
 263 states, and it follows from the reconstruction bound in Section 5.2: a report
 264 from that site is a readout of a code that the two interventions leave identical.

265 A three-site delay line witnesses failure after one round and exact report-

ing after two; cutting the path prevents delivery at every round. A report
 may still happen to be correct about one fixed world. The bound concerns
 responsiveness across interventions.

Throughout this section *prediction* names a reconstruction map, and the
 relevant family of states, the metric, the tolerance, the communication graph
 and the deadline are all declared inputs, as are the represented variables and
 the fidelity counted as accurate. Reading the map as temporal neural pre-
 diction would require dynamics; reading the resulting code count as physical
 memory capacity would require a hardware model, which a state-dependent
 readout would have to extend.

6 What the conditions constrain in a digital candidate

The results of Section 2 hold for any candidate, and for a large language
 model (LLM) running on graphics processing units (GPUs) four of them are
 checkable rather than merely stateable. Those four are where this framework
 bears on claims made for such systems. It does not exclude them, and no re-
 sult here could; what it does is fix what a claim of unity or self-representation
 commits its maker to measuring.

A quantitative bound for a linear bottleneck. Let a finite spatial
 coupling operator K have orthonormal eigenmodes e_i with $|\lambda_1| \geq \dots \geq |\lambda_N|$.

286 Every linear approximation A of rank at most d satisfies

$$\sum_{i=1}^N \|Ke_i - Ae_i\|^2 \geq \sum_{i=d+1}^N \lambda_i^2. \quad (11)$$

287 Retaining the first d modes attains equality. A linear encoder and readout
 288 factoring through d real coordinates therefore pay this error. Applying the
 289 bound requires a spectrum for the target and a rank constraint on the actual
 290 approximation. Query/key dimension alone supplies no such constraint on
 291 softmax attention: even a two-position rank-one logit matrix can yield rank-
 292 two attention. The result prices an explicitly linear bottleneck in a finite
 293 representation of a spatial kernel; it gives no universal gap between trans-
 294 formers and continuous fields. Supplemental Material derives the bound and
 295 checks the softmax counterexample.

296 A fixed synthetic task makes the scope visible. Two local views share
 297 scene and self coordinates; the target is the global state and a self-relative
 298 coordinate. On the parity-held-out family, a rank-two linear decoder’s input-
 299 weighted predicted squared-error floor is 2.000, and its held-out error is 2.000.
 300 On a separate scene-family shift, the full linear decoder fits training inputs
 301 to 0.000 error but has 2.667 held-out error. This is failure of training-family
 302 coverage, not a violation of the spectral theorem. A small causal atten-
 303 tion decoder and a coupled phase code also lose held-out accuracy on that
 304 shift; neither is governed by the linear rank floor. The measured code, over-
 305 lap perturbations, self-state interventions and communication deadlines are

306 described in Supplemental Material. The comparison supplies no hardware-
307 energy or experiential claim.

308 **Codes, not reports.** Reconstruction across a declared family requires the
309 encoder to distinguish states more than 2ε apart when the target error is
310 ε (Eq. (10)). Verbal reports alone do not establish that condition. For
311 a proposed self-model, declare the family and fidelity, identify the internal
312 code under intervention, then test whether it separates the family. A token
313 sequence is one possible code; an activation region is another and requires
314 its own measurement.

315 The same interface has an information ceiling. For any finite joint law
316 of an internal state X and a length- k token sequence Y over vocabulary V ,
317 $I(X; Y) \leq H(Y) \leq k \log_2 |V|$ bits. The proof uses the full joint law and
318 allows dependent emissions and zero-probability outcomes. A binary copy
319 attains one bit per emitted binary symbol; an independent output conveys
320 zero. This counts the declared output sequence, so a cache, timing signal or
321 other accessible state must be included separately when it forms part of the
322 channel.

323 **The region, not the readout.** If the code is a restriction of the can-
324 didate’s state to a region, that region must distinguish the relevant family
325 before a readout is chosen. In one forward pass of a decoder-only transformer,
326 the causal mask restricts position t to earlier positions; an intervention out-
327 side that causal past cannot be reconstructed from its activation alone.

328 Across passes, prior output adds a separate channel. The bounded-fan-in
329 communication result of Section 5 applies to a declared physical intercon-
330 nect and deadline; a full-support attention graph does not satisfy its sparse-
331 neighbourhood premise. Supplemental Material gives the graph bounds and
332 their hardware scope.

333 **Agreement about what.** Unity requires the candidate’s local descriptions
334 to agree on independently identified shared quantities (Section 4). Common
335 timing, a shared residual stream and fluent output supply none of that, for
336 the reason Section 2 gives: coordination bounds content disagreement only
337 through encoders already related on overlaps. Which quantities two parts
338 of a model describe in common is a decoding question, carrying the failure
339 modes of Section 8. The cover is a choice, and choosing it to secure agreement
340 empties the claim.

341 **Physical scope.** Digital representations can satisfy compatibility and re-
342 construction. The coupling-gated route additionally requires a calibrated
343 physical coupling, its noise regime and an independently identified relation
344 from phase order to shared content. An attention matrix or a shared clock
345 supplies none of these; the cortical proposal carries the same physical obliga-
346 tions. Finite-resolution capacity depends on code-space volume, resolution
347 and readout gain, so continuity alone favours no substrate. Supplemental
348 Material gives the capacity and hardware-support comparisons. Neither the
349 installed-energy condition nor those comparisons exclude a digital realization.

350 The conditions impose measurements on a digital candidate but settle
 351 no question of machine experience. Supplemental Material places this com-
 352 parison beside biological naturalism, substrate-dependent computation and
 353 current proposals about LLM organization.

354 7 Thermodynamics and composition

355 The phase and representation results concern coupling, content disagreement
 356 and reconstruction error. Thermodynamics constrains the physical opera-
 357 tions that could implement them, and it does so from the side rather than
 358 along the spine: no edge below derives coherence, compatibility or reconstruc-
 359 tion from a heat bound, and the resource condition these edges terminate in
 360 is necessary without being restrictive for any macroscopic candidate (Sec-
 361 tion 3.2). What this half contributes is a ledger in which a resource claim
 362 is stated in checkable units. Finite-register capacity obeys $H(p) \leq \log |X|$;
 363 erasure heat depends on specified system–bath assumptions (Landauer, 1961;
 364 Norton, 2005; Shenker, 2000). A separate boundary result concerns a field
 365 reaching disconnected vacuum components (Kibble, 1976). The composition
 366 connects these objects through the explicit hypotheses in Table 1.

367 7.1 Probability current supplies a dissipation bound

368 For a positive normalized phase density ρ , diffusion $D > 0$ and current
 369 $J = v\rho - D\partial_\theta\rho$, define $\sigma_J = \int_{-\pi}^\pi J^2/(D\rho) d\theta$. Continuous density and current

370 give the sharp bound

$$\left(\int_{-\pi}^{\pi} J d\theta\right)^2 \leq D\sigma_J. \quad (12)$$

371 A uniform density under constant drive ω attains $\sigma_J = \omega^2/D$. In contrast,
 372 the identical-frequency gradient model has positive self-consistent station-
 373 ary order above $K = 2D$ with $J = 0$ and $\sigma_J = 0$. Thus neither posi-
 374 tive order nor its time average forces positive dissipation: a stationary or-
 375 dered density is already a counterexample. A cortical maintenance-power
 376 bound requires an identified nonequilibrium drive, its probability current
 377 and a thermal conversion; phase order alone does not supply them. The
 378 current functional is distinct from the substrate squared-drift objective used
 379 in the plasticity controls. For a positive normalized periodic density satisfy-
 380 ing the classical continuity equation, the rotation-invariant order magnitude
 381 $r = |\int e^{i\theta} \rho d\theta|$ obeys $\dot{r}^2 \leq D\sigma_J(1 - r^2)$ wherever $0 < r < 1$ and differentia-
 382 tion under the integral is valid. For a smooth path staying in that interval,
 383 $\int_0^\tau \sigma_J dt \geq [\arcsin r(\tau) - \arcsin r(0)]^2/(D\tau)$. This gives a positive current-cost
 384 floor for a specified finite change in order, even though a stationary ordered
 385 density can have zero cost. A finite-volume onset test with a declared order
 386 threshold exceeds the floor; its threshold was selected after exploratory runs,
 387 and its cost is a probability-current account rather than thermal or metabolic
 388 power. It does not bound an externally chosen $\dot{K}$: the uniform density is an
 389 exact time-dependent solution of zero current cost for every coupling sched-
 390 ule, including positive schedules with any prescribed derivative.

391 7.2 Feedback changes the information constraint

392 For passive memory X_t driven by an environmental signal S_t , the predictive
393 bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (13)$$

394 The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
395 et al., 2012). Zero retained information also satisfies it, so a task and a
396 learning rule require additional specification. Relating this bound to Lan-
397 dauer heat requires a common thermal scale and budget, and memory waste
398 bounded by erased entropy.

399 An acting agent changes the signal it will receive. Specify policy, world,
400 observation and memory-update channels. For their finite laws,

$$\begin{aligned} I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\ &\quad - I(X_t; S_{t+1} \mid S_t). \end{aligned} \quad (14)$$

401 The second conditional term vanishes in the passive model. With feedback,
402 the signed difference can be negative as action creates correlations with the
403 future. A causal reading requires an environmental state sufficient to describe
404 the relevant dynamics.

405 For a controlled substep, the active thermodynamic bound instead uses
406 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
407 $\Delta H(X, S) + \langle Q \rangle / (k_B T) \geq 0$ under local detailed balance, with thermal scale
408 $k_B T > 0$ and heat positive into the reservoir (Horowitz and Esposito, 2014;

409 Ito and Sagawa, 2013). Every forward path of positive probability must have
410 a positive-probability reverse for this finite identity. A missing reverse gives
411 infinite divergence. The supplement develops the support conditions and
412 composes actuation with memory updates on their shared intermediate law.

413 **7.3 Learning and repeated operation need explicit re-** 414 **sources**

415 The finite models in Supplemental Material separate policy selection, learning
416 from a phase-derived sensor and repeated operation. A binary closed-loop
417 witness lets a register install the phase energy landscape, lets the phase evolve
418 under a bath-supported channel, senses the new phase and updates that same
419 register. A fixed decoder trained on one family predicts the next phase on
420 a held-out family with accuracy 0.783 under an informative sensor, versus
421 0.552 under a matched blind sensor. Their marginal phase order is the same.
422 The executed law also charges installation, sensing, reset and preparation
423 against a finite work store; phase-flip bath heat is recorded separately. The
424 symbol task and mode prices are supplied toy inputs, so this does not identify
425 a cortical mechanism or a heat cost for the full sensing and reset protocol.

426 For that model, common energy observable $\mathcal{E}$, thermal scale $k_B T$, local
427 detailed balance and positive support at each stage give

$$k_B T [H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle \mathcal{E} \rangle_0 - \langle \mathcal{E} \rangle_N \leq B + \langle \mathcal{E} \rangle_0 - \langle \mathcal{E} \rangle_N, \quad (15)$$

where expected cumulative work stays within the supplied allowance B at every prefix. This bounds expectations. Separate constructions track a store on each trajectory, include finite sources, charge sensor clearing and prepend preparation. They specify when a mean allowance represents an available resource; an expected budget alone permits overdrawn paths.

For the declared process, a finite source with positive net cost funds finitely many cycles. A common physical ledger for the sensing and reset channels, and a cortical calibration of every mode price, remain separate obligations.

7.4 What the composition establishes

The passive and feedback composition theorems each accept eight connecting implications (Table 1). In the active branch, a named substep at the register's thermal scale satisfies

$$k_B T > 0, \quad k_B T [H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (16)$$

with B the register's heat. The upper allocation follows from a ledger whose operations deliver that total heat to one reservoir and whose other operations do not increase joint entropy. Those ledger assumptions are physical inputs. Neither Landauer's lower bound nor microscopic reversibility supplies them. The regime in which those premises are exhibited is a near-equilibrium one: strictly positive forward and reverse elementary paths, local detailed bal-

447 ance at one thermal scale, and entropy differences and heats of order $k_B T$,
 448 the witnesses being two-bit systems with interaction energies of that size.
 449 Cortex has not been shown to satisfy these premises. The illustrative cor-
 450 tical metabolic expenditure over one region and one phase-diffusion time is
 451 2.4×10^{14} times kT at body temperature, but identifying it with the register
 452 heat budget B requires a common physical ledger. Reading the register at
 453 the scale where its premises are physical, such as a single channel, requires
 454 E45Active to carry that step’s budget to the convergence of a continuum
 455 kernel. The active branch is an alternative route to the same node rather
 456 than a stronger one, and no result of Section 3.2 passes through it: the
 457 installed-energy condition is derived from E56 and E67.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy conse-
 quence. The branches share E12, E23 and E56–E89. Exact overlap compatibility
 and the encoding–readout structure are additional inputs. Supplemental Material,
 Table S1, maps the node-level results.

Connection	Status	Required assumption
E12 <i>capacity and boundary</i>	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23 <i>boundary to register</i>	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.

Table 1 (continued)

Connection	Status	Required assumption
E34 / E34Active <i>register to dissipation</i>	Bridge assumption	Passive: predictive memory at the register's thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations' compressiveness. Both are exhibited at thermal-scale heats on finite state spaces with positive reverse paths, which is not the regime a cortical register occupies.
E45 / E45Active <i>couplings to continuum</i>	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Convergence follows from mesh refinement rather than from the budget, which selects no regime at macroscopic scale. Refinement and physical identifications remain inputs.
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.

Table 1 (continued)

Connection	Status	Required assumption
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours. Where the cover's nerve is connected the same edge carries through the sites its patches share, at a residual linear in that nerve's diameter; where the configuration is frequency-locked it carries through the coupling graph's algebraic connectivity λ_2 instead of through any patch count, which asks the cover for nothing and is declared hardware data in the standing of κ .
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $\Lambda < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 3.2's occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

Both branches also take a thermodynamic cover carrying exact overlap compatibility and a reflexive boundary specifying encoding and read-out. They conclude that the glued section is the unique fixed point of self-

461 prediction. Compatibility and accuracy at that section enter as premises;
462 uniqueness follows. Identifying encoding with restriction needs the separate
463 resonance condition of Eq. (8).

464 Each of the eight edges is supplied rather than derived from the one above
465 it. E12 is equivalent to its conclusion, so capacity and boundary are inde-
466 pendent roots rather than a step from one to the other; both E45 discharges
467 take their convergence from refinement data and not from the thermody-
468 namic bound; E78 states that coherence relaxes the supplied cover, and the
469 coherence-to-content estimate of Eq. (6) bounds disagreement only for en-
470 coders already shared on overlaps. The inferential content of the composition
471 is its last step, where contraction and a fixed glued section give uniqueness.
472 What composing them establishes is that the premises are consistent, named
473 in one place and machine-checked together, and not that coherence follows
474 from installed energy or self-representation from coherence.

475 Joint witnesses combine a register, boundary, refining mesh and small
476 spatial cover; active witnesses add controlled thermal updates and a priced
477 installation. They establish satisfiability under their stated identifications.
478 Mesh convergence follows from refinement data, and the witnesses establish
479 no common cortical mechanism. Supplemental Material maps the claims to
480 their Lean declarations and records the counterexamples. The development
481 declares no additional axioms beyond its libraries' foundational axioms.

482 8 What the observations establish

483 The composition holds under its premises. Whether observation can estab-
484 lish those premises is a separate question, and the three requirements it
485 connects concern different objects: phase dynamics, local descriptions and
486 internal encodings. Their observables must retain these distinctions. The
487 following controls show which mechanisms remain indistinguishable, which
488 temporal approximations fail, which states one observable cannot separate,
489 and what independent content measurements must establish. Each is a finite
490 numerical regime on a declared model, with full parameters and controls in
491 Supplemental Material, and none of them calibrates a cortical quantity.

492 8.1 Stationary distributions leave the mechanism un- 493 determined

494 Equation (2) gives a concentration–coherence relation with no fitted shape
495 parameter. Estimate r from circular resultant length and a from density
496 shape: inverting R from the measured r would make agreement automatic.
497 Even independently estimated agreement identifies a statistical family shared
498 by different mechanisms.

499 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
500 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
501 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
502 oscillators with prescribed drift $-h(t)\sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,

503 and diffusion D_0 have that same density. Their stationary distributions co-
504 incide window by window despite different causes.

505 Resolved dynamics supply additional information. In the ideal model with
506 continuous paths, quadratic variation accumulates at rate $2D$ and conditional
507 increment means constrain drift. Separating endogenous coupling from a
508 matched common drive additionally requires a measured drive or controlled
509 perturbation; identifying electromagnetic coupling requires evidence specific
510 to that interaction. Supplemental Material gives the equivalence construction
511 and its numerical check.

512 **8.2 Finite ramps change the observable onset**

513 The square-root law in Eq. (3) describes stationary solutions. A changing
514 coupling requires time for the phase distribution to approach those solutions.
515 In a deterministic mean-field control that crosses the threshold symmetrically,
516 none of the scanned rates meets the declared tracking tolerance: slowing
517 the crossing does not remove the terminal residual under that initialization.
518 A slow control variable therefore supplies no general guarantee of branch
519 tracking at onset. Supplemental Material gives the residual, rate range and
520 admissible rates away from threshold.

521 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
522 at 8 coupling speeds from $v = 0.1$ to $v = 0.0001$. Escape is the first of three
523 consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay
524 escape (Fig. 3); only 5/32 replicas escape within the fastest window. The

525 6 uncensored legs supply the delay scaling below; the decade-spaced ones
526 among them have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted
527 onset exponents differ from those of the exact stationary branch evaluated
528 on the same grids. At the fastest speed the ensemble mean never reaches the
529 fit window, so its fitted slope does not estimate stationary onset.

530 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
531 makes the apparent exponent depend on ramp speed and fitting window.
532 The specified deviation-based critical ramp speed is not bracketed by this
533 sweep. With fluctuating coupling, the relevant stationary comparison also
534 depends on rate: the mean coupling approximates order in the tested fast
535 regime, while averaging stationary order over coupling works for the tested
536 slow drives confined to one side of threshold. Slow crossing drives can satisfy
537 neither approximation.

538 **The prediction.** Over the 6 uncensored legs the mean delay follows $\Delta K \propto$
539 $v^{0.444}$. The interval from fitting those means is $[0.394, 0.494]$; resampling
540 replicas independently within each speed gives a 95% interval $[0.398, 0.496]$,
541 with fraction 0.019 of 10000 bootstrap fits at or above $1/2$. Thus this finite
542 sweep does not measure the ideal square-root exponent. A deterministic
543 mean-field run seeded at the fluctuation floor gives exponent 0.425 with the
544 same estimator and escape criterion; tightening its criterion to $r \geq 0.05$ gives
545 0.490. These controls show that the criterion affects the fitted exponent, but
546 do not prove that it accounts for the entire finite-population shortfall.

547 The ideal delayed-onset calculation motivates a square-root scaling hy-
 548 pothesis; its exponent is generic to suitable bifurcations and does not iden-
 549 tify the coupling medium. A within-subject emergence design can vary the
 550 propofol taper and test how an operational onset changes with that manipu-
 551 lation, but interpreting the result as a coupling-rate law requires a measured
 552 or calibrated relation between taper and $\dot{K}$. The escape criterion must be
 553 set relative to each recording's pre-threshold fluctuation scale. Section 8.1
 554 describes how D could be estimated from resolved phase traces.

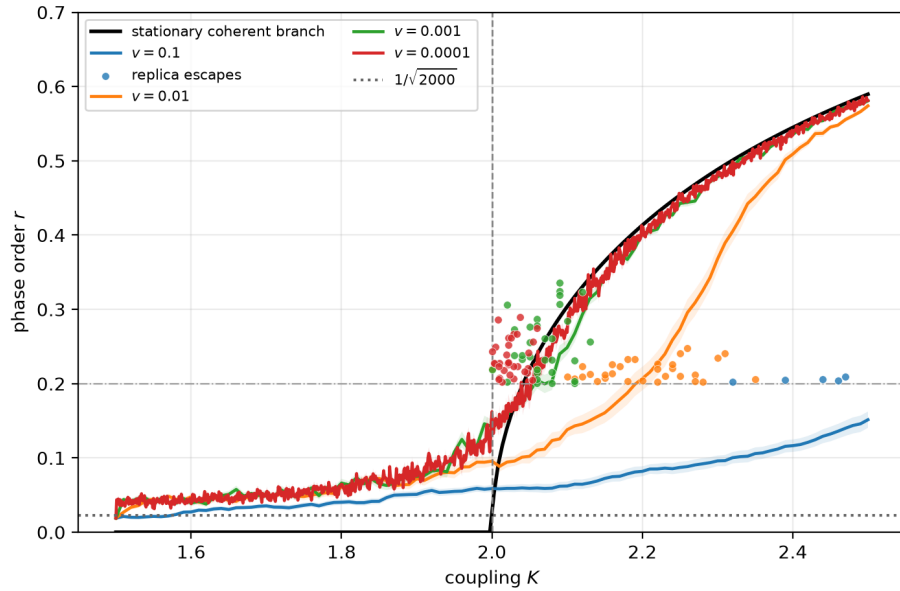


Figure 3: Finite-rate departures from the stationary branch. Curves show ensemble-mean order for four ramp speeds v , with one-standard-error bands. Coloured points mark each replica's first sustained escape, plotted at its recorded order; censored replicas have no point. The dashed vertical line is the static threshold, the dash-dotted horizontal line is the escape criterion, and the dotted horizontal line is the finite-size reference. The replica-bootstrap interval for the delay exponent is reported in the text.

555 Additional controls address integration bias, population size and spatial
556 or signed coupling; the refined signed-network crossing bracket $K_{\text{eff}}/D \in$
557 $[1.958, 2.009]$ contains 2.

558 **8.3 A winding state leaves the order parameter am-** 559 **biguous**

560 The global order parameter r discards spatial arrangement. On a periodic
561 sheet, a phase field can wind once around the sheet and have $r = 0$ while
562 neighbouring sites remain locked. At identical frequencies, every such wind-
563 ing is stationary for an isotropic symmetric kernel: contributions at opposite
564 separations cancel. It violates the quarter-turn initial-spread premise of the
565 convergence result, which is why that result can still select the uniform state.

566 In the spatial control, the winding persists below the interpolated de-
567 cay length 0.1871 mm and unwinds above it (Fig. 4). At 0.1 mm, uniform
568 and winding starts under the same coupling, diffusion and kernel end at
569 $r = 0.9657$ and $r = 0.0083$; patch-local order is 0.9015. Under quenched fre-
570 quency spread 1.5 rad s^{-1} , the measured retention boundary is 0.1421 mm,
571 and the corresponding order comparison is 0.9418 against 0.0348 over 3 frozen
572 realizations. These are finite controls, not a cortical wave identification or a
573 derived transition.

574 Linearization about a winding reweights each separation's coupling by
575 the cosine of its phase advance. The resulting symmetric kernel has explicit

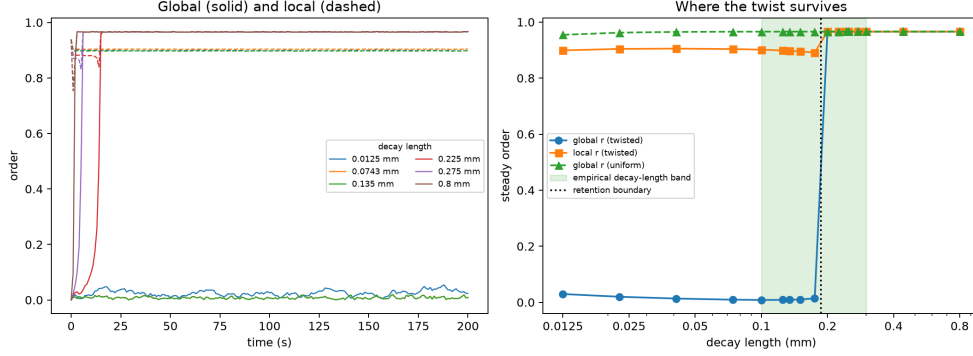


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

576 mode rates: nonnegative reweighted coupling gives no growing mode, while
 577 a positive rate destabilizes that winding. The criterion concerns linear rates,
 578 not asymptotic stability, and the measured retention boundary need not
 579 equal its limit. An amplitude extension adds an existence condition: $\Gamma + \lambda >$
 580 0, where Γ is linear gain and $\lambda = \sum_d f_d(\cos \Delta\theta_d - 1)$ comes from the same
 581 kernel. For the nonnegative kernels used here, a broad kernel can remove a
 582 winding from that model even though the phase-only equations retain it as
 583 a stationary state.

584 Neither model turns these states into measured cortical spirals. The pe-
 585 riodic sheet cannot carry a lone phase defect, the relaxational dynamics has

no propagation speed, and the amplitude variable has no calibrated relation to a recorded envelope. Supplemental Material gives the spectral, amplitude and defect results. Locally, Eq. (6) can still bound content disagreement on a winding patch when shared encoders are supplied, although its global $r = 0$ form is uninformative. A nonzero ring winding number q also obstructs a global real-valued phase lift; it has no identified content meaning here. Thus an onset criterion based only on r can register a change of basin as well as a change of coupling. The protocol of Section 9.3 needs a spatial phase observable.

8.4 Coherent adaptation can lose environmental alignment

An adaptive coupling rule might supply representational structure, but its objective must reward the relevant agreement. For symmetric coupling, pairwise sine terms cancel in the population mean drift. Minimizing squared drift at that fixed mean therefore rewards equalizing drift across nodes. In the frequency-cluster control, this favors coupling between unlike-frequency clusters, whereas the diagnostic environmental template rewards coupling within clusters.

The formalization proves descent of the squared-drift objective for a specified gradient flow at fixed phases. Its functional differs from thermodynamic entropy production: at a coherent stationary density the probability current

607 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
608 rial gives the operator, its gradient and the conditions of the descent result.

609 The finite control co-evolves noisy phases and symmetric, nonnegative
610 couplings at fixed total resource. Its template is a diagnostic covariance
611 of latent signals, never a training input. Across three illustrative seeds,
612 second-half order remains between 0.9167 and 0.9222, while the within-
613 cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–
614 0.6135 (Fig. 5). Random-update, permuted-gradient and frozen-kernel con-
615 trols are reported in Supplemental Material.

616 The immediate post-update objective reduction disappears when aver-
617 aged over complete update intervals. A prespecified grid of 6 rate–cadence
618 combinations contains no case meeting the 5% sustained-reduction criterion
619 while retaining coherence and losing alignment. The finding is coherent adap-
620 tation away from the specified structure, with no demonstrated sustained cost
621 reduction.

622 Kernel alignment and overlap compatibility are different observables.
623 This control addresses acquisition of its designated template under its par-
624 ticular objective. Learning models can explicitly include data in an objective
625 (Corberi et al., 2026); an account of agreement about the cup must specify
626 the content its learning rule is meant to preserve.

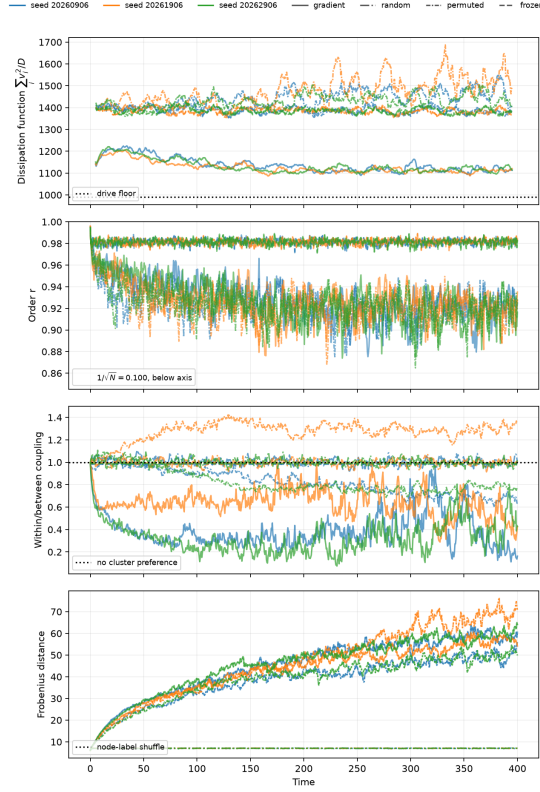


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

627 8.5 Content agreement needs independent decoders

628 A neural test needs independently identified domains, local descriptions and
629 restriction maps. Two decoders should estimate a shared content on over-
630 lapping domains before their agreement is examined. Discrepancy requires a
631 null under decoding noise and controls using deliberately compatible and in-
632 compatible examples. An implementation on synthetic data separates these
633 cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by trial;
634 the compatible construction scores 0.476 on an interval containing one half.
635 Shared-prior shrinkage can nevertheless manufacture agreement, and regional
636 bias can manufacture disagreement. Per-patch decoding diagnostics detect
637 these failures; the compatibility statistic alone does not.

638 Phase agreement cannot replace that measurement. On constructed con-
639 tent, phase carries a share 1.000 of the content entropy when content is built
640 from it and 0.000 when it is not. A phase-derived statistic reports com-
641 patibility on phase-locked patches whose local descriptions disagree. The
642 implemented content statistic detects their incompatibility. These controls
643 validate a measurement on known constructions; a neural content model and
644 decoder remain to be established.

645 Choosing the cover also chooses a candidate subject: trivial descriptions
646 can agree while carrying little content, and a partition can conceal disagree-
647 ment. The useful choice must therefore be constrained by what neural ac-
648 tivity encodes. The exact composition assumes compatibility of its supplied
649 cover. The content mechanisms above identify additional hypotheses under

650 which agreement can arise; they leave the empirical choice of content and
651 the connection to that cover explicit.

652 **8.6 Estimator calibration on exploratory EEG**

653 The concentration range and the dependence structure decide whether the
654 concentration-coherence relation can be tested at all. Under the specified
655 log-density estimator and bias-floor criterion, discrimination at the observed
656 concentration ceiling $a = 0.542$ requires 3000 independent sites in the scanned
657 designs, rising to 31000 when sites share a mean in groups of 10. These
658 counts describe that estimator and dependence model; the intended recording
659 requires its own calibration. Supplemental Material reports the bin-count
660 controls and the full dependence-aware design.

661 That ceiling comes from exercising the estimators on public propofol-
662 sedation EEG (Bajwa et al., 2025), pooling channels in 100-ms win-
663 dows. Bipolar and common-reference montages produce different residuals.
664 Bipolar-montage scalp EEG sits in the linear Bessel regime ($a_{\max} = 0.542$,
665 deviation from $a/2 < 1\%$). The exercise confirms the estimator is well-
666 behaved in this regime but does not test the nonlinear prediction. The
667 bipolar results are compatible with the Bessel relation and with monotone
668 alternatives that agree to this precision over the sampled range. Temporal
669 pooling changes the estimand, and sedation recordings leave the awakening
670 hypothesis untested.

671 An informative dataset therefore needs sufficient concentration range or

precision, calibrated estimators and an observation model for dependence,
source mixing and reference choice. Intracranial recordings (ECoG, sEEG or
LFP) or narrowband burst analysis may reach a more informative concentration range, but that range must be measured. At the observed ceiling, the specified design requires 3000 independent sites or 31000 sites under its clustered dependence model; these are design calculations, not sample requirements transferable to another recording. Public mouse calcium imaging provides thousands of spatially located neurons across waking, sleep and anesthesia (Oomoto et al., 2026), but its 7.65 Hz frame rate is below the phase-estimation band used here. Site count, state coverage and bandwidth are separate requirements.

9 The cortical hypothesis and an awakening test

The cortical proposal identifies a physical source of coupling and asks whether its changes help explain recovery. The tests in Section 8 determine what observations must distinguish before a phase trace can support this identification.

689 9.1 Physical identification and calibration

690 The cortical hypothesis identifies part of effective coupling with endogenous
691 extracellular electromagnetic interactions. Fields can modulate spike timing
692 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
693 propose coordination of distributed neural information (Pinotsis & Miller,
694 2022; McFadden, 2020). Phase organization under propofol motivates study-
695 ing this regime (Bastos et al., 2021), but does not measure coupling.

696 The formal field predicate requires a jointly continuous kernel, a normal-
697 ized substrate, coupling equal to the kernel average, and no single site pro-
698 viding more than half that strength. Toy constructions satisfy it. A cortical
699 realization must specify the scale at which the kernel and its reduction ap-
700 ply: calcium imaging reports different state-dependent modularity patterns
701 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

702 Cortical columns are the obvious candidate unit, and they answer one
703 of the two demands the formalism makes. As the oscillator index they are
704 a serious proposal: they are delineated by response properties and anatomy
705 rather than by the agreement they would be used to establish (Mountcastle,
706 1997), and a phase can be estimated from a column-sized territory at the
707 electrode spacings used for cortical wave measurements (Townsend et al.,
708 2015). As the content cover they are a poor one, for three separate reasons.
709 A parcellation into non-overlapping columns is a disjoint cover, which satisfies
710 the compatibility obligation without constraining anything (Section 4). The
711 cover condition additionally requires the patches to exhaust the substrate,

712 while much of association cortex has no accepted columnar parcellation. And
713 the column's standing as a canonical cortical unit is contested (Horton &
714 Adams, 2005; Rakic, 2008). Offering columns for both roles at once therefore
715 supplies the oscillator index; the content cover remains an open empirical
716 question, which the bridge assumption E78 records.

717 The spatial reduction is also testable. A row-sum aggregation at zero
718 diagonal tracks the uniform-coupling control over the tested spatial decay
719 range, while frequency heterogeneity displaces the operational threshold. The
720 fit window biases both estimates relative to $2D$; agreement with the control
721 is therefore the supported conclusion. Supplemental Material reports the
722 reduction, its numerical error and the distinct assumptions needed for the
723 kernel-average identification in Eq. (5).

724 The field commitment requires physical evidence beyond a choice of math-
725 ematical representation, and the available support comparisons do not supply
726 it. A finite matrix spread over positive-mass cells preserves its total cou-
727 pling and phase correlation exactly, while point-supported vanishing under
728 an atomless measure and a disadvantage from excluding a better-correlated
729 pair concern specified support constraints; Section 6 reads the same compar-
730 isons against a digital candidate. Neither supplies energy prices.

731 A spike-time shift alone supplies no coupling rate. The dimensionless
732 product of population size N , timing sensitivity $\mathcal{T}$, field amplitude E and

733 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N \mathcal{T} E f. \quad (17)$$

734 This rate describes how induced timing shifts accumulate into drift. Nei-
735 ther it nor the installation factor κ follows from a heat bound. Testing the
736 physical threshold requires independent estimates of the interaction kernel,
737 its aggregation and diffusion in the same cortical regime.

738 **9.2 Extracellular geometry and recovery**

739 The proposed control variable is extracellular geometry. Extracellular vol-
740 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
741 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
742 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
743 at $0.3\text{--}0.6 \text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
744 an effective-medium description predicts that a larger extracellular fraction
745 raises conductivity and reduces field amplitude.

746 The coupling-gated hypothesis consequently predicts that extracellular
747 contraction on awakening raises E and, holding the other factors in Eq. (17)
748 fixed, raises effective coupling. This argument supplies a direction; the
749 geometry–coupling relation, its magnitude and its role in human recovery
750 require measurement.

751 For a smooth crossing at nonzero speed, evaluating the stationary branch

752 along the changing control gives a deterministic tracking approximation that
 753 fails at onset in the tested crossing control:

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (18)$$

754 This is a deterministic tracking approximation from Eq. (3) at fixed D ; the
 755 crossing controls in Section 8.2 show that it fails as an observed-onset ap-
 756 proximation under the tested initialization. A separation of geometry and
 757 phase time scales can support tracking away from threshold. The awakening
 758 hypothesis needs a calibrated dynamical prediction for an operational onset
 759 above the finite-size fluctuation floor. Crossing K_c alone does not supply a
 760 recovery latency.

761 **9.3 A protocol for the awakening experiment**

762 Neural inertia describes a behavioural barrier and hysteresis between anaes-
 763 thetic induction and emergence (Friedman et al., 2010); recovery can pass
 764 through metastable activity states (Hudson et al., 2014; Proekt & Hudson,
 765 2018). The $v^{1/2}$ exponent of an ideal delayed bifurcation is generic (Baer
 766 et al., 1989; Berglund & Gentz, 2002) and cannot alone discriminate a field
 767 mechanism from a synaptic one. A possible discriminator is spatial onset
 768 relative to independently estimated local field coupling and network connec-
 769 tivity. The present mean-field model has no spatially varying κ and derives
 770 no nucleation order, so a field-based onset map needs a separate model spec-

771 ified before the observations are inspected.

772 For that comparison, partition the recording into prespecified regions i
773 and measure pre-onset order $r_i(t_0)$ and diffusion D_i . Define a field score
774 $h_i = N_i \mathcal{T}_i E_i f_i$ from independent pre-onset field and timing measurements,
775 and a connectivity score c_i from a separately estimated structural network.
776 For the same measured emergence drive $g(t)$, compare $K_i^F(t) = a_F g(t) h_i$ with
777 $K_i^C(t) = a_C g(t) c_i$. Each model has one fitted scale. The local first-harmonic
778 approximation predicts

$$\hat{r}_i^M(t) = r_i(t_0) \exp\left(\int_{t_0}^t [K_i^M(s)/2 - D_i] ds\right), \quad M \in \{F, C\}. \quad (19)$$

779 Predicted onset is the first crossing of a region-specific threshold fixed from
780 pre-onset fluctuations. Fit the two scales on training sessions, then com-
781 pare held-out onset times and site order under the same observation model.
782 Larger field scores predict earlier onset only when baseline order and diffu-
783 sion do not reverse the ranking. This declared approximation omits interre-
784 gional coupling and saturation; a successful ranking would support its field
785 covariate, not prove ephaptic causation. Supplemental Material specifies the
786 controls and failure conditions.

787 A within-subject protocol can manipulate emergence rate, with spatially
788 resolved recordings to compare the proposed field and connectivity models:

- 789 **1. Manipulate emergence rate.** Vary the propofol infusion taper
790 across sessions within the same subject, recording drug concentrations

791 or a validated pharmacokinetic proxy to quantify the manipulation.

792 **2. Measure behavioural recovery.** Record time-to-response to a stan-
793 dard stimulus as the independent behavioural endpoint, rather than
794 estimating ΔK directly.

795 **3. Resolve spatial onset.** Identify anatomical locations where coher-
796 ence first crosses the baseline-scaled threshold and score the two maps
797 of Eq. (19) on held-out sessions. The two covariate maps are fixed
798 before the outcome sessions are inspected.

799 **4. Match physical controls.** Confirm that the observed coherence-
800 to-amplitude relation holds under the specific local diffusion D and
801 observation noise, estimable from the quadratic variation of resolved
802 phase traces.

803 Predeclare the recording window, regional fluctuation thresholds, and a
804 shared timing-error model. Fit the two scales and that observation model
805 on training sessions only; on held-out sessions, score the observed site-onset
806 times and sites that never cross by the same regional predictive rule. Report
807 site order as a secondary outcome and behavioural time-to-response sepa-
808 rately. Sessions without a crossing are censored observations, not zero delays
809 or exclusions. Compare the two scores only where independently estimated
810 field and connectivity maps predict distinguishable onset patterns at the
811 recording resolution. A held-out field-model advantage would support that
812 specified spatial predictor; it would not identify an ephaptic cause, shared

813 content or experience. Supplemental Material gives the controls and pilot
814 quantities needed to judge whether the comparison is informative.

815 **10 Discussion**

816 **10.1 Unity and a minimal self as interpretive motiva-** 817 **tion**

818 We propose the state glued from compatible local descriptions as a formal
819 correlate of conscious unity, and its accurate internal reconstruction as a cor-
820 relate of a minimal self. In the cup example, this would mean one description
821 that preserves the cup’s relation to the perceiver as well as the scene. Com-
822 patibility permits distinct contents and uncertainty; the shared quantities
823 must be chosen from independent evidence about what each region encodes.
824 The reconstruction need not be a verbal or autobiographical account. Its
825 proposed relation to experienced perspective remains an interpretation. Nei-
826 ther identification is among the eight hypotheses in Table 1, and satisfying
827 them would not measure experience.

828 **10.2 What a test of the full proposal must establish**

829 A test needs separate estimates of effective coupling and diffusion, shared
830 content on overlapping regions, and reconstruction from a specified encoding
831 at a declared fidelity. The cover, relevant state family and decoder must

832 be fixed without selecting them for agreement. An independent experiential
833 endpoint is needed to evaluate the interpretation; absence of a report alone
834 does not define absence of experience. A unified episode with incompati-
835 ble decoded descriptions would challenge the unity identification. A unified
836 episode lacking the proposed self-encoding would challenge the minimal-self
837 claim. Conversely, satisfying the mathematical conditions without such an
838 episode would challenge their sufficiency as correlates.

839 For the physical route, cortical actuator modes and prices, the geometry-
840 coupling relation and a dynamical onset model remain to be measured. The
841 field/connectivity comparison in Section 9.3 evaluates one spatial onset pre-
842 diction. Controlled changes in coordination and independently decoded con-
843 tent agreement would address the representational connection. The synthetic
844 decoder and EEG exercises calibrate parts of these measurements; they do
845 not identify a neural content code or test experience.

846 **10.3 Explanatory scope**

847 Integrated-information, global-workspace and predictive-processing accounts
848 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
849 1997; Dehaene et al., 2014; Friston, 2010). This account specifies conditional
850 relations among coupling, local content and reconstruction. Its formal field
851 condition requires a continuous kernel, a normalized substrate, a specified
852 average coupling and noise, and a site-contribution bound. Those conditions
853 do not identify an electromagnetic substrate; the extracellular interpretation

needs the separate calibration of Section 9. The joint witness shows that the formal hypotheses can hold together in a declared model. A cortical realization and an experiential interpretation require additional evidence.

The remaining explanatory question is whether these relations occur in neural activity and matter for experience. The framework identifies candidate correlates and tests of their physical and representational premises. It does not explain why satisfying them would be accompanied by experience.

Notation

Table 2: Symbols used throughout, with the section that defines each. Where one letter carries more than one meaning, the meanings are separated by the section they are read in.

Symbol	Defined	Meaning
θ_i, ω_i	Eq. (1)	Phase and natural frequency of oscillator i
N	Eq. (1)	Number of oscillators in the population the quantity is read on
D	§2, Supp.	Phase diffusion in inverse time; phase spread $D \leq \pi/2$
K_{ij}, K	Eq. (1)	Pairwise and scalar mean-field coupling, $K_{ij} = K/N$
K_c	Eq. (2)	Synchronization threshold, $K_c = 2D$
r, r_N	Eq. (1)	Resultant length (order parameter), on the population it is read on
a, ψ	Eq. (2)	Concentration $a = Kr/D$ and collective phase of the stationary density
q	Section 8.3	Winding number of a phase field read around a ring
μ	Eq. (4)	Substrate measure, normalized to one
Ξ, ϕ_i, u_i, c_i	Eq. (4)	Actuator configuration; mode profile, installation price and occupancy of mode i

Table 2 (continued)

Symbol	Defined	Meaning
U_{inst}, κ	Eq. (5)	Installed energy and the mode-data conversion factor bounding $K_{\text{mf}} \leq \kappa U_{\text{inst}}$
L	Section 4.2	Lipschitz constant of an encoder or readout
ε, δ	Section 4.2	Content and observation discrepancy tolerances
M, b	Eq. (10)	Size of a declared family of states and code length in bits
Λ, τ	Eq. (8)	Contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ and its relaxation time
T	§3.2, §5	Temperature in the thermal scale $k_B T$; number of communication rounds to a deadline
ν	Section 6	Fan-in of a communication graph: the most sites any one site hears from
$k_B T, B$	Eq. (16)	Thermal scale, read as one symbol, and heat budget of a register
$\gamma, \mathcal{T}, f$	Eq. (17)	Field-to-coupling conversion rate, timing sensitivity and frequency
α	Section 9	Extracellular volume fraction
λ	§8.3, §9	Eigenvalue the coupling kernel supplies at a winding; extracellular tortuosity
Ω, λ_2	Section 4.2	Common rotation rate of a frequency-locked configuration, and the algebraic connectivity of the coupling graph
β, k	Section 4.2	Chord diameter of a patch and the diameter of a cover's nerve, in hops
$\mathcal{C}, e_{i,x}$	Section 4.2	Content space and the phase encoder patch i reads at site x
Φ, R	Eq. (10)	Encoder and readout of the reconstruction criterion
$\mathcal{A}, F$	Eq. (8)	Encoding region and the reconstruction map $F(s) = \text{readout}(s _{\mathcal{A}})$
$\mathcal{E}$	Eq. (15)	Energy observable shared by the stages of a continuing protocol
z_i, Γ, A_{ij}	Section 8.3	Complex amplitude, linear gain and coupling matrix of the amplitude layer
$\Delta\theta_d$	Section 8.3	Phase a winding advances across separation d

Table 2 (continued)

Symbol	Defined	Meaning
E	Eq. (17)	Field amplitude, in the cortical identification
U	§3.2	Stored energy $U(\Xi)$ of a configuration
s, s_*	§4–5	A global section and the one the reconstruction fixes

862 Data and code availability

863 The Lean development, simulation scripts, tests and saved numerical
864 summaries are maintained in the GitHub repository `bobaseb/coupling_`
865 `gated_cortical_coherence`. The formalization uses Lean 4.34.0-rc2 and
866 Mathlib commit `5ce203ecd53f8c3e47ee5530d7c86b94a55f0183`. Publi-
867 cation simulation values are generated from saved summaries without re-
868 running production sweeps. The exploratory EEG analysis uses Open-
869 Neuro dataset `ds005620` (Bajwa et al., 2025) (subjects 1010, 1016, 1022,
870 1033, 1045, 1056, resting awake and sedation runs 1–2); preprocessing
871 and selection are described in Supplemental Material and implemented in
872 `simulations/empirical_collapse.py`. The repository contains the repro-
873 ducible analysis and does not redistribute the raw recordings.

874 Funding and competing interests

875 This research received no external funding. The author declares no compet-
876 ing interests.

877 Use of AI tools

878 AI assistance to this work was substantial and is not separable by task. Sev-
879 eral coding assistants and chat interfaces were used in an interleaved way
880 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
881 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
882 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
883 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,
884 accessed with the Hermes CLI. Between them they contributed to the Lean
885 development, the simulation and analysis code, and the drafting and revision
886 of both documents. Because the tools were used in combination and in alter-
887 nation, no section, proof, figure or result is attributable to any one of them,
888 and none is claimed to be. The author is responsible for all content, including
889 everything a tool produced. Machine checking establishes the stated Lean
890 results under their hypotheses; it does not validate biological interpretations
891 or replace scientific review of generated code and prose.

892 References

- 893 Pinotsis, D. A., & Miller, E. K. (2022). Beyond dimension reduction: Stable
894 electric fields emerge from and allow representational drift. *NeuroImage*,
895 253, 119058.
- 896 Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated informa-

tion theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7), 450-461.

Baars, B. J. (1997). In the Theater of Consciousness: The Workspace of the Mind. *Oxford University Press*.

Dehaene, S., Charles, L., King, J. R., & Marti, S. (2014). Toward a computational theory of conscious processing. *Current Opinion in Neurobiology*, 25, 76-84.

Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature Reviews Neuroscience*, 11(2), 127-138.

Milinkovic, B., & Aru, J. (2026). On biological and artificial consciousness: A case for biological computationalism. *Neuroscience & Biobehavioral Reviews*, 181, 106524. doi:10.1016/j.neubiorev.2025.106524.

Aru, J., Larkum, M. E., & Shine, J. M. (2023). The feasibility of artificial consciousness through the lens of neuroscience. *Trends in Neurosciences*, 46(12), 1008–1017. doi:10.1016/j.tins.2023.09.009.

Seth, A. K. (2026). Conscious artificial intelligence and biological naturalism. *Behavioral and Brain Sciences*, 49, e315. doi:10.1017/S0140525X25000032.

Seth, A. K. (2026). The stuff matters: Consciousness, computation, and biology. *Behavioral and Brain Sciences*, 49, e366. doi:10.1017/S0140525X26106906.

- 917 Gurnee, W., et al. (2026). Verbalizable Representations Form a
918 Global Workspace in Language Models. *Transformer Circuits Thread*.
919 arXiv:2607.15495.
- 920 Aguilera, M., Millidge, B., Tschantz, A., & Buckley, C. L. (2022). How par-
921 ticular is the physics of the free energy principle?. *Physics of Life Reviews*,
922 40, 24-50.
- 923 Biehl, M., Pollock, F. A., & Kanai, R. (2021). A technical critique of some
924 parts of the free energy principle. *Entropy*, 23(3), 293.
- 925 Bruineberg, J., Dolega, K., Dewhurst, J., & Baltieri, M. (2022). The Em-
926 peror’s New Markov Blankets. *Behavioral and Brain Sciences*, 45, e183.
- 927 Pinotsis, D. A., & Miller, E. K. (2023). In vivo ephaptic coupling allows
928 memory network formation. *Cerebral Cortex*, 33(17), 9877-9895.
- 929 Pinotsis, D. A., Fridman, G., & Miller, E. K. (2023). Cytoelectric coupling:
930 Electric fields sculpt neural activity and ”tune” the brain’s infrastructure.
931 *Progress in Neurobiology*, 226, 102465.
- 932 McFadden, J. (2002). The conscious electromagnetic information (cemi) field
933 theory: the hard problem made easy?. *Journal of Consciousness Studies*,
934 9(8), 45-60.
- 935 Pockett, S. (2012). The electromagnetic field theory of consciousness: a
936 testable hypothesis about the characteristics of conscious as opposed to
937 non-conscious fields. *Journal of Consciousness Studies*, 19(11-12), 191-223.

- 938 McFadden, J. (2020). Integrating information in the brain's EM field: the
 939 cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1),
 940 niaa016.
- 941 Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may
 942 guide neocortical network activity. *Neuron*, 67(1), 129-143.
- 943 Anastassiou, C. A., & Koch, C. (2015). Ephaptic coupling to endogenous
 944 electric field activity: why bother?. *Current Opinion in Neurobiology*, 31,
 945 95-103.
- 946 Pockett, S. (2002). Difficulties with the electromagnetic field theory of con-
 947 sciousness. *Journal of Consciousness Studies*, 9(4), 51-56.
- 948 Vöröslakos, M., et al. (2018). Direct effects of transcranial electric stimulation
 949 on brain circuits in rats and humans. *Nature Communications*, 9, 483.
- 950 Verkhratsky, A., & Nedergaard, M. (2018). Physiology of Astroglia. *Physio-*
 951 *logical Reviews*, 98(1), 239-389.
- 952 Mountcastle, V. B. (1997). The columnar organization of the neocortex.
 953 *Brain*, 120(4), 701-722. doi:10.1093/brain/120.4.701.
- 954 Horton, J. C., & Adams, D. L. (2005). The cortical column: a structure
 955 without a function. *Philosophical Transactions of the Royal Society B*,
 956 360(1456), 837-862. doi:10.1098/rstb.2005.1623.

- 957 Rakic, P. (2008). Confusing cortical columns. *Proceedings of the National*
958 *Academy of Sciences*, 105(34), 12099-12100. doi:10.1073/pnas.0807271105.
- 959 Bastos, A. M., Donoghue, J. A., Brincat, S. L., Mahnke, M., Yanar, J., Cor-
960 rea, J., ... & Miller, E. K. (2021). Neural effects of propofol-induced uncon-
961 sciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824.
- 962 Bajwa, I. J., Nilsen, A. S., Skukies, R., Aamodt, A., Ernst, G., Storm, J.
963 F., & Juel, B. E. (2025). A repeated awakening study exploring the capac-
964 ity of complexity measures to capture dreaming during propofol sedation.
965 *Scientific Reports*, 15, 32746.
- 966 Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiyagarajan, M., ... &
967 Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult
968 brain. *Science*, 342(6156), 373-377.
- 969 Syková, E., & Nicholson, C. (2008). Diffusion in brain extracellular space.
970 *Physiological Reviews*, 88(4), 1277-1340.
- 971 Barbour, B. (2017). Analysis of claims that the brain extracellular impedance
972 is high and non-resistive. *Biophysical Journal*, 113(7), 1636-1638.
- 973 Logothetis, N. K., Kayser, C., & Oeltermann, A. (2007). In vivo measure-
974 ment of cortical impedance spectrum in monkeys: implications for signal
975 propagation. *Neuron*, 55(5), 809-823. doi:10.1016/j.neuron.2007.07.027.
- 976 Gabriel, S., Lau, R. W., & Gabriel, C. (1996). The dielectric properties of bio-
977 logical tissues: II. Measurements in the frequency range 10 Hz to 20 GHz.

- 978 *Physics in Medicine and Biology*, 41(11), 2251–2269. doi:10.1088/0031-
979 9155/41/11/002.
- 980 Norton, J. D. (2005). Eaters of the lotus: Landauer’s principle and the return
981 of Maxwell’s demon. *Studies in History and Philosophy of Science Part B:
982 Studies in History and Philosophy of Modern Physics*, 36(2), 375–411.
- 983 Landauer, R. (1961). Irreversibility and heat generation in the computing
984 process. *IBM Journal of Research and Development*, 5(3), 183–191.
- 985 Kibble, T. W. B. (1976). Topology of cosmic domains and strings. *Journal
986 of Physics A: Mathematical and General*, 9(8), 1387–1398.
- 987 Sakaguchi, H. (1988). Cooperative phenomena in coupled oscillator systems
988 under external fields. *Progress of Theoretical Physics*, 79(1), 39–46.
- 989 Strogatz, S. H., & Mirollo, R. E. (1991). Stability of incoherence in a popu-
990 lation of coupled oscillators. *Journal of Statistical Physics*, 63, 613–635.
- 991 Acebrón, J. A., Bonilla, L. L., Pérez Vicente, C. J., Ritort, F., &
992 Spigler, R. (2005). The Kuramoto model: A simple paradigm for
993 synchronization phenomena. *Reviews of Modern Physics*, 77, 137–185.
994 doi:10.1103/RevModPhys.77.137.
- 995 Ott, E., & Antonsen, T. M. (2008). Low dimensional behavior of
996 large systems of globally coupled oscillators. *Chaos*, 18, 037113.
997 doi:10.1063/1.2930766.

998 Dörfler, F., & Bullo, F. (2014). Synchronization in complex net-
 999 works of phase oscillators: A survey. *Automatica*, 50, 1539–1564.
 1000 doi:10.1016/j.automatica.2014.04.012.

1001 Breakspear, M. (2017). Dynamic models of large-scale brain activity. *Nature*
 1002 *Neuroscience*, 20, 340–352. doi:10.1038/nn.4497.

1003 Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics
 1004 of prediction. *Physical Review Letters*, 109(12), 120604.

1005 Kawai, R., Parrondo, J. M. R., & Van den Broeck, C. (2007). Dissipation:
 1006 The phase-space perspective. *Physical Review Letters*, 98(8), 080602.

1007 Horowitz, J. M., & Esposito, M. (2014). Thermodynamics with continuous
 1008 information flow. *Physical Review X*, 4, 031015. doi:10.1103/PhysRevX.4.
 1009 031015.

1010 Ito, S., & Sagawa, T. (2013). Information thermodynamics on
 1011 causal networks. *Physical Review Letters*, 111, 180603. doi:10.1103/
 1012 PhysRevLett.111.180603.

1013 Corberi, F., dello Russo, S., Messuti, G., Scarpetta, S., & Smaldone, L.
 1014 (2026). Thermodynamic learning. *arXiv preprint*. arXiv:2609.04732.

1015 Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur ap-
 1016 plication aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-
 1017 181.

- 1018 Shenker, O. R. (2000). Logic and entropy. *PhilSci-Archive preprint*. philsci-
1019 archive.pitt.edu/115.
- 1020 Rosas, F. E., Mediano, P. A. M., Luppi, A. I., Boyd, A. B., Berta, M., Jensen,
1021 H. J., Gastpar, M., & Carroll, S. M. (2026). Four faces of information in
1022 natural systems. *Nature Reviews Physics*. doi:10.1038/s42254-026-00971-4.
- 1023 Sznitman, A.-S. (1991). Topics in propagation of chaos. In *École d'Été de*
1024 *Probabilités de Saint-Flour XIX—1989*, Lecture Notes in Mathematics,
1025 vol. 1464, pp. 165–251. Springer. doi:10.1007/BFb0085169.
- 1026 Dai Pra, P., & den Hollander, F. (1996). McKean–Vlasov limit for interacting
1027 random processes in random media. *Journal of Statistical Physics*, 84, 735–
1028 772. doi:10.1007/BF02179656.
- 1029 Bertini, L., Giacomin, G., & Pakdaman, K. (2010). Dynamical aspects of
1030 mean field plane rotators and the Kuramoto model. *Journal of Statistical*
1031 *Physics*, 138, 270–290. doi:10.1007/s10955-009-9908-9.
- 1032 Berglund, N., & Gentz, B. (2002). Pathwise description of dynamic pitchfork
1033 bifurcations with additive noise. *Probability Theory and Related Fields*,
1034 122, 341–388. doi:10.1007/s004400100174.
- 1035 Abramsky, S., & Brandenburger, A. (2011). The sheaf-theoretic structure
1036 of non-locality and contextuality. *New Journal of Physics*, 13, 113036.
1037 doi:10.1088/1367-2630/13/11/113036.

1038 Townsend, R. G., Solomon, S. S., Chen, S. C., Pietersen, A. N. J., Martin,
1039 P. R., Solomon, S. G., & Gong, P. (2015). Emergence of complex wave
1040 patterns in primate cerebral cortex. *The Journal of Neuroscience*, 35(11),
1041 4657-4662. doi:10.1523/JNEUROSCI.4509-14.2015.

1042 Xu, Y., Long, X., Feng, J., & Gong, P. (2023). Interacting spiral wave pat-
1043 terns underlie complex brain dynamics and are related to cognitive pro-
1044 cessing. *Nature Human Behaviour*, 7(7), 1196-1215. doi:10.1038/s41562-
1045 023-01626-5.

1046 Nicoletti, G., Di Terlizzi, I., & Busiello, D. M. (2026). Balancing information
1047 and dissipation with partially observed fluctuating signals. *Physical Review*
1048 *Letters*, 137(12), 127101. doi:10.1103/8gx2-rbns.

1049 Kiyooka, D., Oomoto, I., Kitazono, J., Saito, Y., Kobayashi, M., Matsub-
1050 ara, C., Kobayashi, K., Murayama, M., & Oizumi, M. (2026). Single-
1051 cell resolution functional networks during unconsciousness are segre-
1052 gated into spatially intermixed modules. *Cell Reports*, 45(2), 116902.
1053 doi:10.1016/j.celrep.2025.116902.

1054 Oomoto, I., Kiyooka, D., Oizumi, M., & Murayama, M. (2026). Multi-area
1055 single-cell calcium imaging dataset of the mouse cortex across wakefulness,
1056 sleep, and anesthesia. *Scientific Data*. doi:10.1038/s41597-026-08202-2.

1057 Attwell, D., & Laughlin, S. B. (2001). An energy budget for signaling in the

1058 grey matter of the brain. *Journal of Cerebral Blood Flow & Metabolism*,
1059 21(10), 1133-1145. doi:10.1097/00004647-200110000-00001.

1060 Harris, J. J., Jolivet, R., & Attwell, D. (2012). Synaptic energy use and
1061 supply. *Neuron*, 75(5), 762-777. doi:10.1016/j.neuron.2012.08.019.

1062 Nicholls, D. G., & Ferguson, S. J. (2013). *Bioenergetics 4*. Academic Press.

1063 Baer, S. M., Erneux, T., & Rinzel, J. (1989). The slow passage through a
1064 Hopf bifurcation: delay, memory effects, and resonance. *SIAM Journal on*
1065 *Applied Mathematics*, 49(1), 55–71. doi:10.1137/0149003.

1066 Friedman, E. B., Sun, Y., Moore, J. T., Hung, H.-T., Meng, Q. C., Perera, P.,
1067 Joiner, W. J., Thomas, S. A., Eckenhoff, R. G., Sehgal, A., & Kelz, M. B.
1068 (2010). A conserved behavioral state barrier impedes transitions between
1069 anesthetic-induced unconsciousness and wakefulness: evidence for neural
1070 inertia. *PLoS ONE*, 5(7), e11903. doi:10.1371/journal.pone.0011903.

1071 Hudson, A. E., Calderon, D. P., Pfaff, D. W., & Proekt, A. (2014). Recovery
1072 of consciousness is mediated by a network of discrete metastable activity
1073 states. *Proceedings of the National Academy of Sciences*, 111(25), 9283–
1074 9288. doi:10.1073/pnas.1408296111.

1075 Proekt, A., & Hudson, A. E. (2018). A stochastic basis for neural inertia
1076 in emergence from general anaesthesia. *British Journal of Anaesthesia*,
1077 121(1), 86–94. doi:10.1016/j.bja.2018.02.035.